

AHMED ALI

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EDUCATION

University of Miami

Ph.D. Industrial and Systems Engineering

Coral Gables, USA

Aug 2024 - Now

- CGPA: 4.0
- Relevant courses: Machine Learning, Applied Data Analytics, Reinforcement Learning and Decision Making Under Uncertainty, Linear Programming and Extensions, Game and Information Theory, Design of Experiments, and Advanced Production Systems.

Egypt-Japan University of Science and Technology

Master of Science in Industrial Engineering and Systems Management

Alexandria, Egypt

Nov 2021 - May 2024

- CGPA: 3.9
- Relevant courses: Introduction to Industrial Engineering, Operations Research, Advanced Soft Computing, Applied Simulation Modeling and Analysis, and Project Planning and Management.
- Thesis: A Multi-Stage Optimization Approach for External Trucks and Yard Cranes Scheduling in Container Terminals.

Benha University

Bachelor of Science in Production Engineering.

Benha, Egypt

Sep 2014 - Jun 2019

- Cumulative average of 79.3% and general grade Excellent with degree of Honors.
- Relevant courses: Production Engineering, Integrated Design, Operations Research, Engineering Economy, Modeling and Analysis of Industrial Systems, Industrial Relations, Computer in Planning, analysis and Control.
- Graduation Project: "Process Improvement of Electrical Panels Assembly". Lean six sigma concept was utilized to standardize the working procedure for each component, and reduce the number of defective parts.

EXPERIENCE

University of Miami

Research Scholar and Teaching Assistant

Miami, USA

Aug 2024 - Now

- Conducting research on consumer attitudes toward load shifting in electricity usage. The project models a market with a single electricity provider and multiple consumers to evaluate behavioral impacts on demand-side management.

Egypt-Japan University of Science and Technology

Research Scholar

Alexandria, Egypt

Nov 2021 - May 2024

- Working on a research project which aims to improve truck appointment systems in container terminals by simultaneously minimizing the number of empty trips, the deviation from the trucking companies preferred arrival time, and the truck turnaround time.

RESEARCH INTERESTS

Subject areas	Operations research, Mathematical Modeling and Optimization, Machine Learning and Data Analytics.
Applications	Energy, Logistics, Healthcare.

PUBLICATIONS

- A. Talaat, J. Iijima, M.Gheith, and A. Eltawil, "An Integrated k-Means Clustering and Bi-Objective Optimization Approach for External Trucks Scheduling in Container Terminals*," *ICIEA-PapersOnLine*, vol. 35, no. 10, pp. 805–814, 2023, doi: [10.3233/ATDE230109](https://doi.org/10.3233/ATDE230109).

- Talaat, A.; Gheith, M.; Eltawil, A. A Multi-Stage Approach for External Trucks and Yard Cranes Scheduling with CO2 Emissions Considerations in Container Terminals. Logistics 2023, 7, 87. doi: [10.3390/logistics7040087](https://doi.org/10.3390/logistics7040087).

TEACHING EXPERIENCE

University of Miami Miami, USA
Teaching Assistant, Department of Industrial and Systems Engineering Aug 2019 - Now

- Responsibilities: tutoring several courses as well as developing assignments, and grading quizzes and assignments.

Benha University Benha, Egypt
Teaching Assistant, Department of Production Engineering Sep 2019 - Nov 2021

- Responsibilities: tutoring several courses as well as developing assignments, and grading quizzes and assignments.

TAUGHT COURSES

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|--|------------------------------|
| • Computer Applications in Industrial Engineering. | • Engineering Management. |
| • Simulation Modeling and Systems Analysis. | • Design of Machine Element. |
| • Engineering Economy. | • Theory of Machines. |
| • Quality Control. | |

SKILLS

Technical skills	Mathematical modeling, Simulation, Machine learning and Data analysis.
Programming languages	Python, R, and LaTeX.
Software packages	PyTorch, Seaborn, Gurobi solver, Numpy, Pandas, Scikit-learn, Matplotlib, SciPy, Amplpy.
Simulation	Arena, EnergyPlus, AnyLogic.
Languages	Fluent in Arabic and English (C1)

PROFESSIONAL CERTIFICATIONS

- Energy Efficiency Practitioner (EEP), Association of Energy Engineers (AEE) Issued: June 2025 — Certification ID: 574 — Miami, FL, USA

AWARDS AND HONORS

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|---|----------|
| • Outstanding graduate student award for teaching, Industrial and Systems Engineering Department, University of Miami | Apr 2026 |
| • Master's degree scholarship, Egyptian Ministry of Higher Education. | Nov 2021 |
| • Distinct team award, 2019 SRRIC, Xian University, China. | Sep 2019 |

INTERNSHIPS

ALDAWLIA-International Engineering Company 6th October, Egypt
Process Engineer Intern Mar 2019 - July 2019

- Developed and implemented a new approach to minimize the total number of required labors for a family of parts of electrical panels.
- Developed process flow charts in order to identify the process bottlenecks and enhance the performance.

Technological & Electrical System Company Cairo, Egypt
CNC Machining Intern July 2017 - Sep 2016

- Developed codes and identified the suitable operating settings for different types of CNC machines to minimize the machining time and maximize the efficiency.

ADDITIONAL ACTIVITIES

Attended conferences

- The Institute of Industrial and Systems Engineers (IISE) annual conference & Expo — May 2026 *Texas, USA*
- The Institute for Operations Research and the Management Sciences (INFORMS) annual meeting — October 2025 *Georgia, USA*
- 10th International Conference on Industrial Engineering and Applications — April 2023 *Phuket Island, Thailand*

Courses

- Operations, Supply Chain and Systems Engineering, American University in Cairo, May 2018

Competitions

- The 3rd Silk Road Robotics Innovations Competition (2019 SRRIC), Xi'an – China